Mass

1 Convert the following:

a $60\ \text{g}$ to milligrams (mg)

b $0.47\ \text{g}$ to milligrams (mg)

c $87\ \text{mg}$ to grams (g)

d $8505\ \text{mg}$ to grams (g)

e $946\ \text{g}$ to kilograms (kg)

f $6.251\ \text{kg}$ to grams (g)

g $3\frac{3}{5}\ \text{kg}$ to grams (g)

h $8500\ \text{kg}$ to tonnes (t)

i $630\ \text{kg}$ to tonnes (t)

j $9.272\ \text{t}$ to kilograms (kg)

k $0.493\ \text{t}$ to kilograms (kg)

2 Valerie has fruit everyday for breakfast. Today she consumed 45 cherries, which weighed a total of 180 grams. On average, how many milligrams does one cherry weigh?

3 True or False: $5\ \text{kg} > 5600\ \text{g}$

4 What metric unit represents 'one thousandth of a kilogram'?

5 Aoife's bag weighs $\frac{1}{2}$ of a kilogram. How many grams does the bag weigh?

6 The cost of mining gold for a mining company is $\$840$ per gram. How much will it cost the company to mine 0.2 kilograms of gold?

Length

7 State whether the following units can be used to measure length:

a Litres, L

b Centimetres, cm

c Cubic centimetres, cm^3

d Kilometres, km

e Square meters, m^2

f Millimetres, mm

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8 Convert the following lengths, writing your answer as a decimal:

a $80\ \text{mm}$ to centimetres (cm)

b $59\ \text{mm}$ to centimetres (cm)

c $4\ \text{m}$ to centimetres (cm)

d $9.36\ \text{m}$ to centimetres (cm)

e $6\frac{3}{4}\ \text{m}$ to centimetres (cm)

f $9000\ \text{m}$ to kilometres (km)

g $4832.1\ \text{m}$ to kilometres (km)

h $6.52\ \text{m}$ to millimetres (mm)

i $49.1\ \text{cm}$ to metres (m)

j $1402\ \text{cm}$ to metres (m)

k $7.66\ \text{km}$ to metres (m)

l $\frac{4}{5}\ \text{km}$ to metres (m)

9 A skyscraper is 961 metres tall. Convert the height of the building to:

a Kilometres

b Centimetres

10 Neville is $141\ \text{cm}$ tall and Rosey is $1.78\ \text{m}$ tall.

a Convert Rosey's height from metres to centimetres.

b Who is shorter, Rosey or Neville?

11 Plant 1 grows $4.8\ \text{cm}$ in a week, while Plant 2 grows $50\ \text{mm}$ in the same week. Which plant grows more in the week?

Area

12 Convert the following:

a $3\ \text{m}^2$ to square centimetres (cm^2)

b $7\ \text{m}^2$ to square centimetres (cm^2)

c $0.56\ \text{km}^2$ to square metres (m^2)

d $54\,800\ \text{m}^2$ to hectares (ha)

e $810\ \text{mm}^2$ to square centimetres (cm^2)

f $10.4\ \text{ha}$ to square metres (m^2)

g $34\,000\ \text{cm}^2$ to square metres (m^2)

h $5.73\ \text{cm}^2$ to square millimetres (mm^2)

13 Convert the following:

a $790\,000\,000\ \text{cm}^2$ to square kilometres

b $6\ \text{ha}$ to square metres

c $20\,000\ \text{m}^2$ to hectares

d $8\,750\ \text{m}^2$ to hectares

14 Find the area of a square of side $290\ \text{cm}$, in square metres.

15 A park has an area measuring 19 acres. Given that 1 acre is approximately $4000\ \text{m}^2$, find the area of the park in square metres.

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16 A property covers an area of $4.5\ \text{ha}$. Given that $1\ \text{ha}$ is $10\,000\ \text{m}^2$, calculate the following:

- **a** The area of the property in square metres.
- **b** The area of the property in acres, given that 1 acre is approximately $4000\ \text{m}^2$.

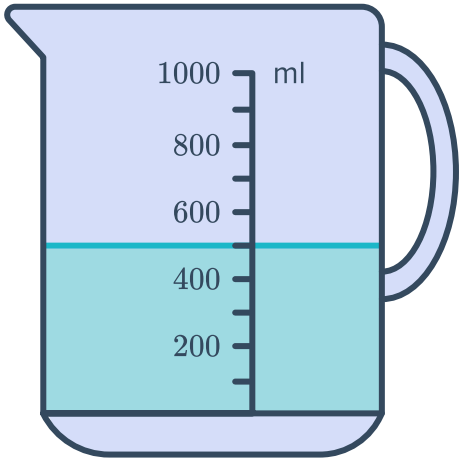
17 A large cattle station covers an area of 320 acres. Given that 1 acre is approximately $0.4\ \text{ha}$, and $1\ \text{ha}$ is $10\,000\ \text{m}^2$, calculate the following:

- **a** The area of the cattle station in hectares.
- **b** The area of the cattle station in square kilometres.

Capacity

18 What metric unit represents 'one thousandth of a litre'?

19 How much water is in the jug shown? Give your answer in litres.



20 Convert the following:

- **a** $2000\ \text{mL}$ to litres (L)
- **b** $4920\ \text{mL}$ to litres (L)
- **c** $240\ \text{mL}$ to litres (L)
- **d** $5\ \text{kL}$ to litres (L)
- **e** $1.67\ \text{kL}$ to litres (L)
- **f** $0.4\ \text{L}$ to millilitres (mL)
- **g** $1.54\ \text{L}$ to millilitres (mL)
- **h** $3\frac{4}{5}\ \text{L}$ to millilitres (mL)
- **i** $84.6\ \text{L}$ to kilolitres (kL)
- **j** $1425\ \text{L}$ to kilolitres (kL)

21 James drank $\frac{9}{10}$ litres of water. How much did he drink in millilitres?

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23 279 children went on a picnic. Each child drank 400 millilitres of orange juice. Calculate the number of litres consumed altogether.

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